1 Detailed Course Contents

Chapter 1: Introduction to System Design

Fundamentals and design methodology

1 Definition, scope and importance of system design
2 Objectives and principles of good system design
3 High-Level Design (HLD) and Low-Level Design (LLD)
4 Functional and non-functional requirements
5 System architecture, components, modules and interfaces
6 Client-server architecture and request-response lifecycle
7 Design constraints, assumptions and trade-offs
8 System design process and documentation

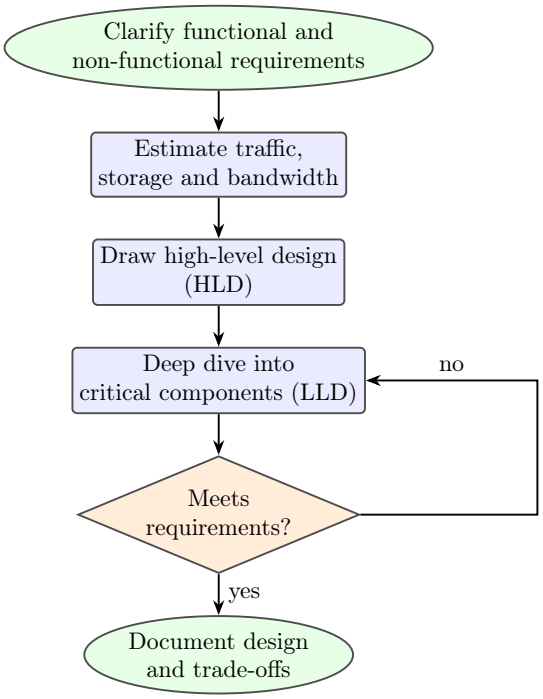


Figure 1: The system design process. Designs loop back to the deep-dive step until the non-functional requirements are met.

Chapter 2: Performance, Scalability and Capacity Planning

Performance analysis and resource estimation

1 Performance versus scalability
2 Latency, throughput and bandwidth
3 Response time and latency percentiles (p50, p95, p99)
4 Vertical and horizontal scaling
5 Availability, reliability and durability
6 Bottlenecks and single points of failure
7 Back-of-the-envelope estimation
8 Traffic, storage, bandwidth and resource estimation
9 Load testing and performance measurement

Chapter 3: Networking and Communication Fundamentals

Network services and request routing

1 OSI and TCP/IP reference models
2 TCP, UDP and connection establishment

3 HTTP/1.1, HTTP/2 and HTTP/3
4 HTTPS, TLS and secure communication
5 Domain Name System (DNS)
6 Forward proxy and reverse proxy
7 Content Delivery Network (CDN): push and pull models
8 Load balancers: Layer 4 and Layer 7
9 Load-balancing algorithms and health checks
10 API gateways and request routing

Chapter 4: Database Design and Data Storage

Data modeling, storage and database scaling

1 Relational and non-relational databases
2 SQL, NoSQL and key-value data models
3 Document, wide-column, graph and relational databases
4 Database schema design and normalization
5 Indexing and query optimization
6 Transactions and ACID properties
7 Isolation levels, locking and concurrency control
8 Database replication and read replicas
9 Sharding and partitioning
10 Database connection pooling
11 Data consistency, backup and recovery

Chapter 5: Caching and Content Delivery

Reducing latency and database load

1 Fundamentals and benefits of caching
2 Client-side, server-side and distributed caching
3 Cache-aside, read-through and write-through
4 Write-back and write-around caching
5 Cache invalidation and expiration policies
6 LRU, LFU and FIFO eviction algorithms
7 Cache consistency and cache stampede
8 Distributed caches and Redis-style data structures
9 CDN caching and cache-control headers
10 Cache performance and common failure scenarios

Chapter 6: API Design and Software Architecture

Interfaces, services and architectural styles

1 RESTful API design principles
2 GraphQL and gRPC
3 API endpoints, HTTP methods and status codes
4 Pagination, filtering, sorting and versioning
5 Idempotency and backward compatibility
6 Monolithic and modular architectures
7 Microservices architecture
8 Service discovery and inter-service communication
9 Event-driven architecture
10 API gateway and Backend-for-Frontend (BFF)
11 Service decomposition and design trade-offs

Chapter 7: Distributed Systems

Coordination, consistency and distributed data

1 Introduction to distributed systems
2 Distributed computing challenges and partial failures
3 CAP theorem and consistency-availability trade-offs
4 Strong, eventual and weak consistency
5 Data replication strategies
6 Leader-follower and multi-leader replication
7 Quorums and read/write consistency
8 Distributed transactions and coordination
9 Consensus concepts and leader election
10 Distributed locking and clock synchronization
11 Consistent hashing and data partitioning

Chapter 8: Messaging, Queues and Asynchronous Processing

Background jobs and event processing

1 Synchronous versus asynchronous communication
2 Message queues and message brokers
3 Producer-consumer architecture
4 Pub-sub and event streaming
5 Background jobs and scheduled tasks
6 Delivery semantics: at-most-once, at-least-once and exactly-once processing considerations
7 Retries, timeouts and dead-letter queues
8 Idempotent consumers and duplicate-message handling
9 Ordering, batching and backpressure
10 Stream processing and change data capture (CDC)

Chapter 9: Availability, Fault Tolerance and Reliability

Building resilient and recoverable systems

1 Availability metrics and service-level objectives
2 Redundancy and elimination of single points of failure
3 Failover and health monitoring
4 Active-active and active-passive architectures
5 Fault tolerance and graceful degradation
6 Circuit breakers and bulkheads
7 Rate limiting and throttling
8 Load shedding and overload protection
9 Disaster recovery, backups and recovery objectives
10 Reliability testing and failure analysis

Chapter 10: Security, Monitoring and Deployment

Production operations and system protection

1 Authentication and authorization
2 Sessions, tokens, JWT and OAuth 2.0 fundamentals
3 Role-based and attribute-based access control
4 Encryption in transit and at rest
5 Secrets management and secure API design
6 Logging, metrics and distributed tracing
7 Observability and alerting
8 Containers and deployment fundamentals

9 CI/CD, configuration management and infrastructure automation
10 Resource utilization, cloud cost and capacity optimization

Chapter 11: Advanced System Design Concepts

Specialized architectures and modern applications (elective)

1 Search engines and full-text indexing
2 Object storage and large-file handling
3 Real-time systems and WebSockets
4 Geospatial systems and location-based services
5 Time-series databases
6 Unique ID generation and distributed counters
7 Recommendation systems and feed generation
8 Vector databases and semantic search
9 Introduction to AI/ML system architecture
10 LLM serving, inference gateways and retrieval-augmented generation (RAG)

Chapter 12: Case Studies and Practical System Design

Application of concepts to real-world systems (practical)

1 URL shortener (TinyURL)
2 Pastebin and file-sharing systems
3 Distributed rate limiter
4 Notification and email delivery system
5 Chat and real-time messaging system
6 Social media news feed
7 Video streaming platform (YouTube)
8 Ride-hailing and location-tracking system
9 E-commerce and ticket-booking system
10 Distributed file storage (Dropbox)
11 End-to-end architecture design, bottleneck analysis and trade-off justification